

PAVAN KUMAR SADASHIV

AI Systems Chief Architect | GPU Systems & High-Performance Runtime Engineer | Founder & MD

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Platform: [hrlpavan.github.io/hrl-international-website-](https://hrlpavan.github.io/hrl-international-website/) | Charter: github.com/hrlpavan/engineering-compensation-blueprint

EXECUTIVE SUMMARY

High-velocity AI Systems Chief Architect and GPU Systems Engineer with deep technical mastery across low-level C++20/CUDA 12.6 high-performance inference runtimes, Paged KV-Cache virtualization, CUDA Graph replay optimization, and discrete combinatorial optimization (NP-hard CSP Backtracking). Proven track record architecting the **NVIDIA RTX-LocalAI Enterprise Execution Suite** for Ada Lovelace and Blackwell GPUs, hardware-accelerated OpenFX visual compute plugins for Hollywood post-production suites (DaVinci Resolve), and zero-bloat native backend microservices. Founder and Managing Director of HRL International Private Limited with verified competencies in corporate grant financing (Startup India / Karnataka ELEVATE), distributed stream processing (JPMorgan Chase certified), and enterprise cybersecurity governance (Deloitte certified). Author of the master technical treatise *Rule Breaking (Volume 4.0): The 100x AI Chief Architect Manifesto*.

CORE TECHNICAL COMPETENCIES

GPU & Low-Level Compute:	C++20/17, NVIDIA CUDA 12.6 Kernel Programming, Apple Metal Shading Language (MSL), CUDA Graphs, Paged Attention & KV-Cache Virtualization, OpenFX API, GPU Memory Optimization, Tensor Cores.
AI/ML & Vision Systems:	PyTorch, ONNX Runtime, Hardware-Aware FP8 (E4M3/E5M2) & INT4-AWQ/GPTQ Quantization, Speculative Decoding Policy Engines, Monocular Neural Depth Estimation (Depth Anything), ACEScC Color Science.
Algorithms & Optimization:	Combinatorial Discrete Optimization, Constraint Satisfaction Problem (CSP) Backtracking Solvers, Dynamic K-Means Clustering (k=3), Bayesian Classifiers, Sliding Window Token Bucket Rate Limiting.
Backend & Systems:	Native Node.js (node:http, node:fs), SQLite 3 (better-sqlite3, ACID Transactions), Zero-Framework Server Architecture, RESTful API Design, Microservices, Multi-Agent AI Orchestration (Google Antigravity SDK).
DevOps & Build Systems:	CMake, Git, GitHub Actions (CI/CD Pipelines), macOS Application Bundling & DMG Packaging, Linux System Administration.

FLAGSHIP ENGINEERING & GPU PROJECTS

1. NVIDIA RTX-LocalAI Enterprise Execution Suite

Technologies: C++20, CUDA 12.6, Paged KV-Cache, CUDA Graphs, Speculative Decoding, INT4-AWQ, FP8, CMake | hrlpavan.github.io/nvidia-project-by-hrl/

- Architected an on-device AI inference engine, memory management subsystem, and benchmarking suite for NVIDIA GeForce RTX (Ada Lovelace / Blackwell) and DGX-class systems.
- Paged KV-Cache Virtualization (src/memory/)**: Engineered a virtual-to-physical block manager (16 tokens/block) achieving **0.17 µs/request allocation latency** with **0.00% memory fragmentation** on 8GB–24GB VRAM.
- CUDA Graph Replay & Continuous Batching (src/runtime/)**: Pre-captured static compute graphs dropping CPU launch latency to **< 5 µs**, paired with iteration-level continuous batching and chunked prefill (512 tokens/chunk).
- Speculative Policy Engine (src/runtime/speculative.cpp)**: Pairs fast draft models (1B) with target evaluators (8B) for **up to 2.2x speedup** at an 88.4% speculative acceptance rate.
- Hardware-Aware Quantization (src/quantization/, src/kernels/)**: Built custom dequantization GEMM/GEMV CUDA kernels for FP8 E4M3/E5M2 and INT4-AWQ (0.65 ms TPOT, 5,014 MB peak VRAM).

2. AI Cinematic Haze — OpenFX & Fusion GPU Plugin for DaVinci Resolve

Technologies: C++17/20, NVIDIA CUDA, Apple Metal (MSL), OpenFX API, Depth Anything AI, ONNX Runtime, ACEScC | github.com/hrlpavan/hrl-international-website-

- Architected a high-throughput, hardware-accelerated OpenFX visual compute plugin integrated into DaVinci Resolve 21 and Blackmagic Fusion.
- Engineered dual hardware-accelerated GPU backends using Apple Metal Shading Language for Apple Silicon and CUDA kernels for NVIDIA GPUs.
- Integrated monocular neural depth estimation (Depth Anything via ONNX Runtime) to calculate physically accurate volumetric optical atmospheric scattering.
- Preserved uncompressed 12-bit DPX sensor dynamic range through strict Hollywood ACEScC color science management without luminance clipping.

FLAGSHIP ENGINEERING & GPU PROJECTS (CONTINUED)

3. InviMatrix — College Examination & Invigilation Management Portal

Technologies: JavaScript (ES6+), Native Node.js (Zero-Framework), SQLite 3, CSP Backtracking, K-Means Clustering | github.com/hrlpavan/exam-invigilation-dsa-app

- Implemented an NP-hard **Constraint Satisfaction Problem (CSP) Backtracking solver** allocating faculty duties, classroom venues, and multi-department exam timetables with zero collisions.
- Built an **AI/ML Smart Batch Planner** with dynamic K-Means clustering (k=3) and a Bayesian lateral-entry classifier to interleave students across physical benches for anti-cheating dispersion.
- Engineered a high-throughput, zero-framework native Node.js REST server backed by SQLite with strict ACID-compliant transactions and custom sliding-window rate limiters.
- Developed an Apple Liquid Glass UI with dynamic pill navigation and client-side HTML5 2D Canvas A4 seating matrix export engines.

4. OmniTransform AI — Intelligent Process Automation Platform

Technologies: Multimodal AI Pipelines, Phonetic NLP, ElevenLabs Voice AI, Google Cloud BigQuery | github.com/hrlpavan/omnitransform-ai-resources

- Architected an enterprise Intelligent Process Automation (IPA) platform featuring multimodal document parsing and phonetic entity extraction.
- Developed a neural speech pipeline leveraging ElevenLabs Multilingual v2 Voice AI paired with Chladni resonance acoustic physics visualizers.
- Engineered multi-agent coordination architectures utilizing Google Antigravity SDK, Cloud Composer (Airflow), and BigQuery analytics.

5. HRL-Lang — Domain-Specific Language for LLM Agent Orchestration

Technologies: Compiler Design, AST Parsers, Formal Language Specification | github.com/hrlpavan/hrl-lang

- Authored the official grammar and syntax specification for HRL-Lang (Spec HRL-PATENT-SPEC-2026-004-LANG) optimized for autonomous multi-agent orchestration.

PROFESSIONAL EXPERIENCE & LEADERSHIP

Founder & Managing Director — HRL International Private Limited (2026 – Present)

- Founded and incorporated an enterprise software and media engineering venture, directing technical architecture and product roadmap.
- Authored comprehensive grant applications and financial allocation models for the **Startup India Seed Fund Scheme (SISFS)** and **Karnataka ELEVATE Grant 2026**.
- Directed media distribution infrastructures achieving **2.5M+ organic impressions** across digital creator channels (@hrlpremiumstudio, @hrlfllix, @hrlfx).
- Implemented corporate governance, trademark registration frameworks, and DMCA intellectual property protection standards.

MASTER PUBLICATIONS & TREATISES

Rule Breaking (Volume 4.0): The 100x AI Chief Architect Manifesto (Master Treatise)

- Authored the comprehensive 2026 foundational document detailing the paradigm shift from manual line-by-line syntax typing to high-order AI Chief Architecture. [Read Master PDF]

EDUCATION & VERIFIED CERTIFICATIONS

Bachelor of Engineering (B.E.) in Computer Science & Engineering (AI & ML)

Sahyadri College of Engineering & Management (SCEM), Mangaluru, Karnataka, India

- **JPMorgan Chase & Co.** — Distributed Streaming & Software Engineering Simulation (Kafka, REST Controller Architecture, Low-Latency Processing)
- **Deloitte Australia** — Cybersecurity Defense & Enterprise Threat Modeling Simulation (Perimeter Defense, Access Control Governance, Incident Response)
- **Blackmagic Design** — DaVinci Resolve Color Science & Visual Computing Integration